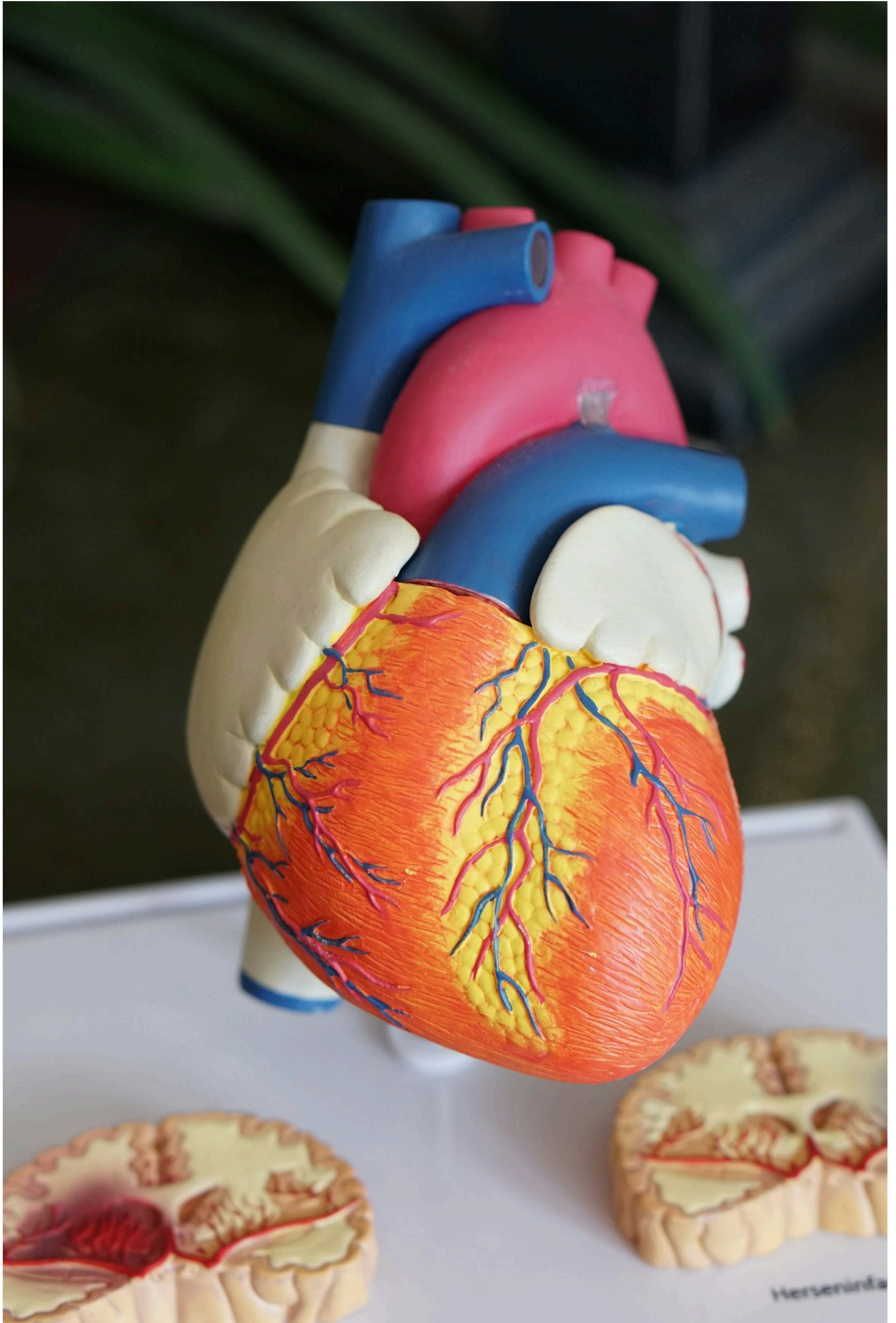


116. THE CYCLIST'S KEY MUSCLE: THE HEART

 knowledgeiswatt.substack.com/p/115-the-cyclists-key-muscle-the-heart

Knowledgeiswatt

November 21, 2025



Endurance performance is largely determined by the power of the aerobic engine. When athletes rely on oxidative pathways to produce energy, they use systems that generate fewer metabolites and delay peripheral fatigue. The main performance indicators that reflect the power of the aerobic system are $\text{VO}_{2\text{max}}$, lactate/ventilatory thresholds, FTP, and Critical Power.

The main determinant of strong aerobic qualities is the **cardiovascular system**, because oxygen is transported from the lungs to the working muscles through the bloodstream. **The king of the cardiovascular system is the heart**— which acts as a pump responsible for delivering blood, and therefore oxygen, to the working muscles. **Because oxygen transport depends on cardiac output, having a bigger and more functional heart is a key determinant of endurance performance.** And, just like any other muscle, the heart's size can differ between different individuals, influencing the aerobic qualities.

How do cardiac dimensions differ among athletes of different performance levels?

A study published by Brown and Colleagues (Liverpool John Moores University, UK) on *Scandinavian Journal of Medicine & Science in Sports* in 2020 tried to answer this question. (1)

WHAT DID THEY DO?

Using conventional and speckle tracking 2D echocardiography, the heart dimension of the 3 following groups was evaluated:

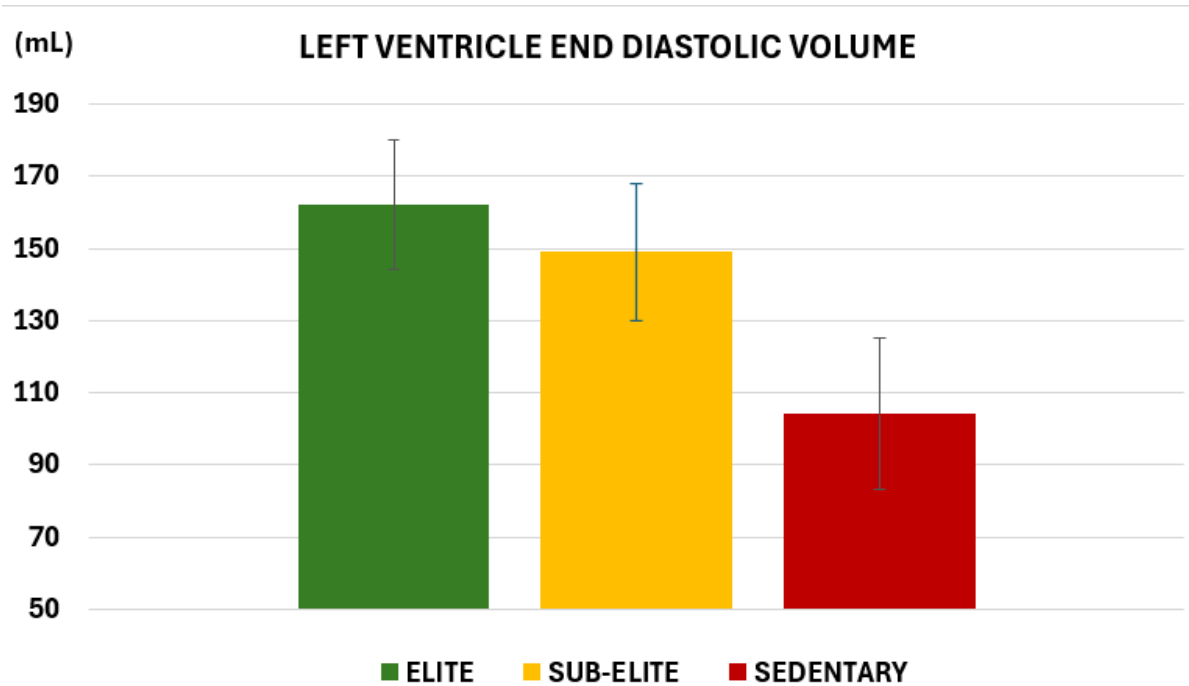
1. 69 male elite cyclists competing in UCI World Tour or UCI Pro Continental level events;
2. 30 sub-elite cyclists actively racing under a 1st, 2nd, or 3rd category British Cycling license;
3. 46 healthy, non-smoking male non-athlete university students/ staff engaging in fewer than 3 hours recreational activity per week;

WHAT DID THEY FIND?

Here some heart size and functionality parameters:

Left Ventricle End Diastolic Volume:

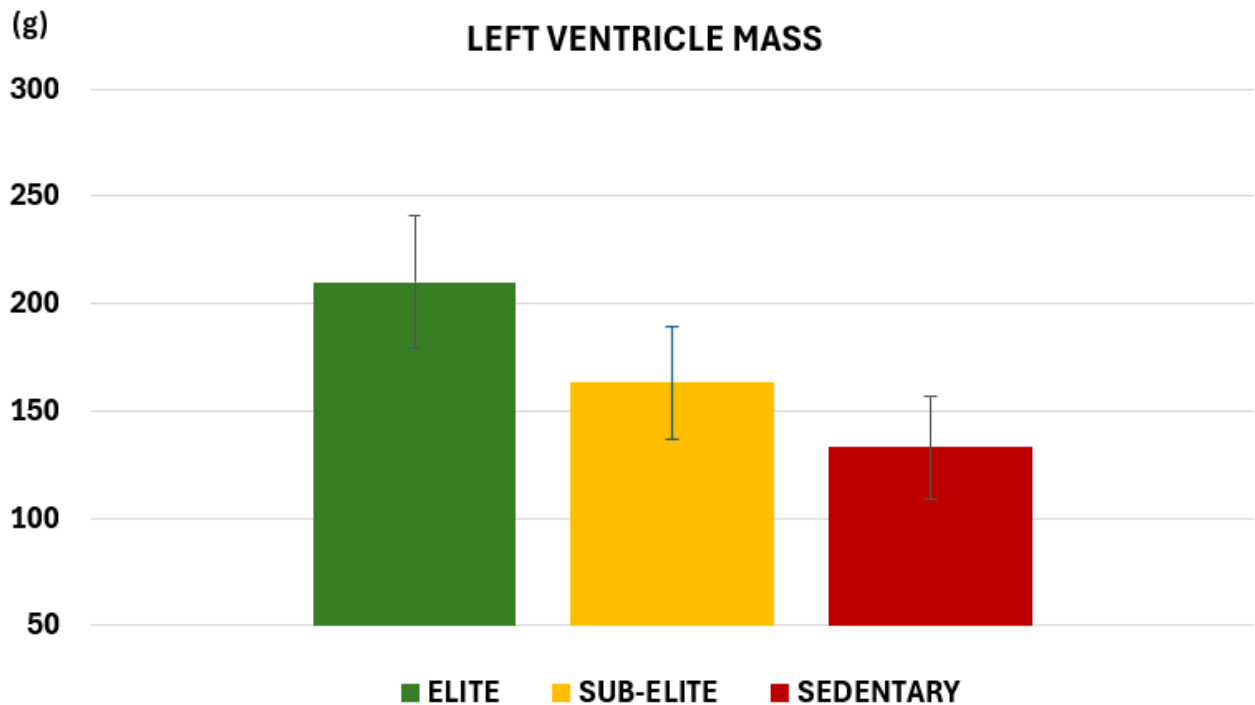
- ELITE: 162 ± 18 ml
- SUB-ELITE: 149 ± 19 ml
- UNTRAINED: 104 ± 21 ml



Data from Brown et al. (1)

Left Ventricle Mass (g):

- ELITE: 210±31 g
- SUB-ELITE: 163±26 g
- UNTRAINED: 133±24 g



Data from Brown et al. (1)

What is the left ventricle?

The heart has four chambers: two upper chambers (atria, right and left) and two lower chambers (ventricles, right and left).

The left ventricle is the chamber where blood collects before being pumped into the systemic circulation, delivering oxygen and nutrients to the working muscles.

For this reason, it can be considered the most important heart's chamber for endurance performance.

What is the end diastolic volume?

End-diastolic volume is the amount of blood in the ventricle at the end of diastole, which is the phase when the heart muscle relaxes and the chamber fills with blood. It represents the maximum volume of blood the ventricle holds before contraction (systole), when blood is pumped into the systemic circulation to go to the working muscles. End-diastolic volume is an important measure because it reflects both the size of the heart chamber and its functional capacity to pump blood efficiently.

So, left ventricular mass reflects the size of the heart, while end-diastolic volume reflects both its size and its functional capacity.

PRACTICAL TAKEAWAYS:

Cardiac size and function are greater in elite athletes than in sub-elite athletes, and greater in sub-elite athletes than in sedentary individuals.

GABRIELE'S COMMENT

There's not much more to add here. **Elite endurance athletes have bigger, stronger hearts than their lower-level peers, and sub-elite cyclists still outperform sedentary individuals.**

These differences align with the variations in aerobic power across these groups, highlighting the main role of **heart and cardiovascular system** for endurance performance.

Is this about genetics or training? Like most traits linked to athletic performance, it's probably **a mix of both**. An interesting insight could come from the exceptionally large chest size of some great cyclists—think **Fausto Coppi** or **Jonas Vingegaard**—suggesting that cardiac size may be at least in part inherited.

We'll see more about this topic in the next KIW Research Notes!

Thanks for reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCE:

1. Brown B, Millar L, Somauroo J, George K, Sharma S, La Gerche A, Forsythe L, Oxborough D. Left ventricular remodeling in elite and sub-elite road cyclists. Scand J Med Sci Sports. 2020 Jul;30(7):1132-1139.



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